FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

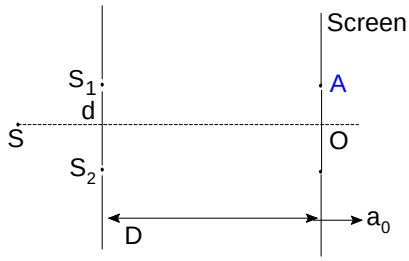
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The pitch of a screw gauge is 0.5 mm and there are 100 division on its circular scale. When there is nothing between jaws, the instrument reads +5 division. In measuring the diameter of a wire, there are 6 division on main scale and 74th division coincide with main scale, then the diameter of wire is

(A) $3.370\ mm$ (B) $6.370\ mm$

(C) $3.345\ mm$ (D) $3.335\ mm$

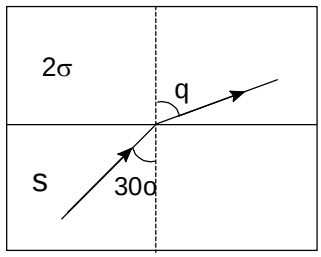
2. In the given YDSE apparatus as shown in figure, wavelength of light used is λ. The initial distance between screen and plane of slits is 'D'. Point A is point on screen where 4th order maxima lies. Now, screen is moved from rest with constant acceleration a_0 towards right. Then find the time after which second order maxima is formed at point A.



(A) $\sqrt{\frac{2D}{a_0}}$ (B) $\sqrt{\frac{D}{a_0}}$

(C) $2\sqrt{\frac{D}{a_0}}$ (D) $\sqrt{\frac{D}{2a_0}}$

3. Figure shows the two conducting medium of electric conductivity σ and 2σ the direction of electric current density are as shown by the arrow. The dotted line shows the normal of the boundary surface. Then, find the value of θ.



(A) $\tan^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{1}{4}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (D) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Space for Rough work

4. Two concentric spherical conducting shell of radius 'R' and Radius '2R' has dielectric medium having dielectric constant K=3 between these. The dielectric strength of dielectric is $E_0 = 8\ Volt/m.$ Then find dielectric breakdown voltage between the shell. (Take $R = \frac{1}{2}m$)

(A) 4 Volt (B) 2 Volt

(C) $\frac{1}{2}$ Volt (D) 0.25 Volt

5. Two spheres of masses 1 kg and 3 kg are on a horizontal smooth surface. Sphere of mass 3 kg is given velocity $V_0 = \frac{8}{3}m/s$ towards other sphere such that it makes an angle of 30° with line of impact. Then, find maximum possible loss of kinetic energy of two sphere during collision.

(A) 8/3 Joule (B) 4 Joule

(C) 2 Joule (D) 2/3 Joule

6. A long straight conductor, having a circular cross section of radius 2m carries a current along its length such its magnetic field varies as $B = Kr^2, 0 \le r \le 2m,$ where K is a positive constant. The slope of the current density (j)vs radial distance 'r' at $r = \frac{1}{2}m$ *is* $\frac{\alpha K}{\mu_0}$. The value of α is.

(A) 4 (B) 8

(C) 2 (D) 3

7. A particle is moving in anticlockwise in a circle under the influence of a force $\vec{F} = \left[\left(x^2 - y\right)\hat{i} + \left(x + 5y - z\right)\hat{j} + \left(3x - 4y + z\right)\hat{k}\right] N$ where x, y and z are in meter. The circle lies in the xy plane with its centre at the origin and has radius of 2m. The work done by the force as the particle completes one revolution is (in Joule)

(A) 4π (B) 8π

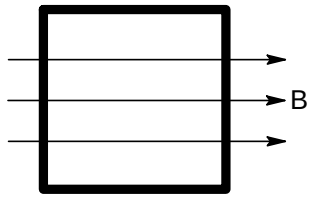
(C) 2π (D) 16π

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

8. A uniform rigid square wire frame of side length 'ℓ' and mass '4m' is placed on smooth horizontal surface where exists uniform horizontal magnetic field of Intensity B. Now, current that is double of minimum current required to flip over frame is switched on in the coil. Then, just after current is switched on



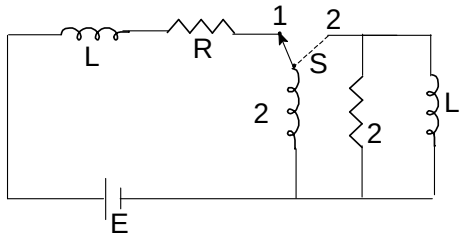
(A) The normal force acting on coil is $\frac{32mg}{5}$

(B) The normal force acting on coil is $\frac{23mg}{20}$

(C) Angular acceleration of coil is $\frac{6g}{5\ell}$

(D) Angular acceleration of coil is $\frac{3g}{10\ell}$

9. In the given circuit, the switch S is connected to point 1 till steady state is achieved. At t = 0; it is shifted to point 2 till steady state is achieved.



(A) The final current in inductor having inductance 2L is $\frac{E}{2R}$.

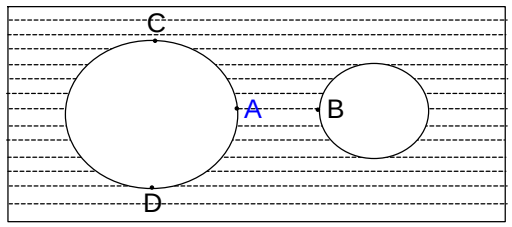
(B) The total amount of heat dissipated through 2R is $\frac{LE^2}{3R^2}$

(C) The total charge which flows through resistor 2R is $\frac{LE}{3R^2}$

(D) The final energy stored in inductor having inductance L is $\frac{2LE^2}{9R^2}$

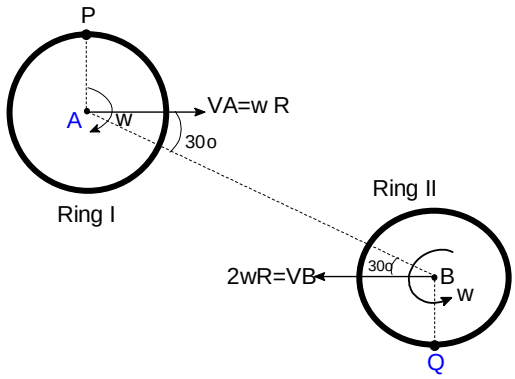
Space for Rough work

10. Figure shows a rectangular plate having two circular holes of radii a and b (a>b). Now plate is heated such that there is temperature rise of ΔT. Then



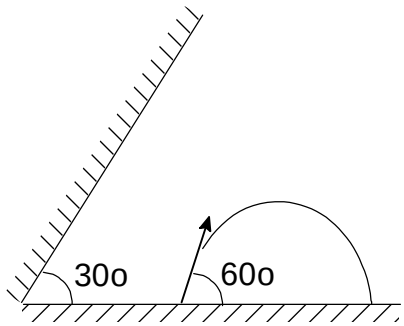
(A) Distance between centre of two holes remain constant
(B) Distance between A and B will increase
(C) Distance between C and D will decrease
(D) Distance between C and D will increase

11. Two identical ring I and II of equal mass 'm' and radius 'R' are moving in opposite direction with linear speed $\omega R \text{ and } 2\omega R$ respectively. Rings are rotating with equal angular speed in opposite direction. The line joining the centres make an angle of 30° with the velocity. Then, at the given instant (AB = 6R)



(A) The angular momentum of ring I about point P is $2mR^2\omega$
(B) The angular momentum of ring I about point Q is $7mR^2\omega$
(C) The angular momentum of ring II about point Q is $2mR^2\omega$
(D) The angular momentum of ring II about point P is $15mR^2\omega$

12. A plane mirror is inclined at an angle 30° with the horizontal surface. A particle is projected from ground (at t = 0) with velocity $V = 20\ m/s$ at an angle of 60° with the horizontal. The image of particle is observed. (The particle does not collide with the mirror). Then



(A) The time when image is at rest with respect to particle is $\frac{2}{\sqrt{3}}$ second
(B) The time when the image is at rest with respect to particle is $\sqrt{3}$ second
(C) Path of image of particle as seen by itself is parabolic
(D) Path of image of particle as seen by itself is straight line

Space for Rough work

13. The energy, the magnitude of linear momentum and the orbital radius of an electron in a hydrogen atom corresponding to the quantum number n are E, P and r respectively. Then according to Bohr's theory of hydrogen atom

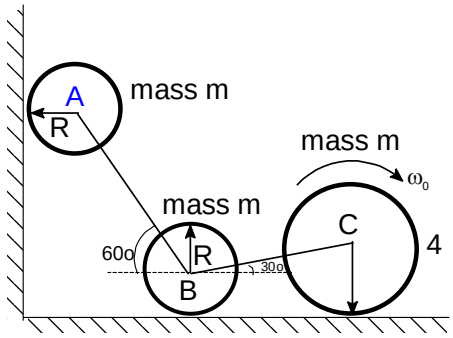
(A) Pr is proportional to n

(B) $E\,\mathrm{Pr}^2$ is proportional to n

(C) $\frac{E}{P}$ is proportional to n^2

(D) $E\,r^2$ is proportional to n^2

14. Two identical disc A and B of equal mass 'm' and radius 'R' are connected by a massless rod. The centres of disc C of mass 'm' and radius 4R and disc B are connected by massless rod which makes an angle of 30^0 with the horizontal. There is no slipping between each disc and surface on which it rolls. When angular speed of disc C is ω_0, rod connecting disc A and B makes an angle 60° with horizontal. Then, at the given instant



(A) Angular speed of disc A is $4\omega_0/\sqrt{3}$

(B) Angular speed of disc A is $2\omega_0/\sqrt{3}$

(C) The total kinetic energy of the system is $28m\omega_0^2R^2$

(D) The total kinetic energy of the system is $26m\omega_0^2R^2$

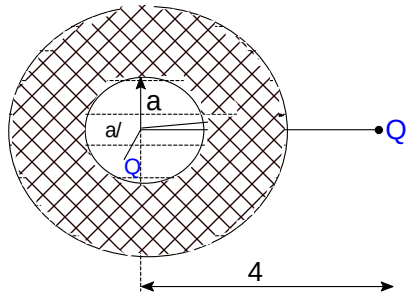
Space for Rough work

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

A positive point charge having charge Q is placed at a distance a/2 from centre of an unchanged spherical conductor of inner radius a and outer radius '2a'. Another positive point charge having charge 'Q' is placed at a distance 4a from centre of a conducting sphere. Then



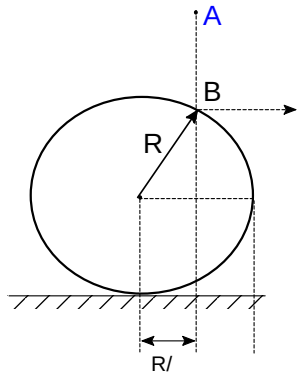
15. Which of the following is incorrect?
(A) The Surface charge density on inner and outer surface of sphere would be non uniform
(B) The Electrostatic potential on outer surface of conductor is $\frac{3KQ}{2a}$
(C) Electric potential at centre of sphere is $\frac{7KQ}{4a}$
(D) Electric field inside cavity is non-uniform.

16. If conductor is earthed. Then, amount of charge which flows through earthed wire is
(A) $\frac{3Q}{2}$ (B) Q
(C) $\frac{Q}{2}$ (D) Zero

Space for Rough work

Paragraph for Question Nos. 17 and 18

A particle of mass 1 kg is released from point A. It collides with sphere of mass 2 kg kept on smooth horizontal surface. After collision velocity of particle becomes horizontal. There is no friction at any point of contact (Given $AB = 20m,\ g = 20\ m/s^2$)

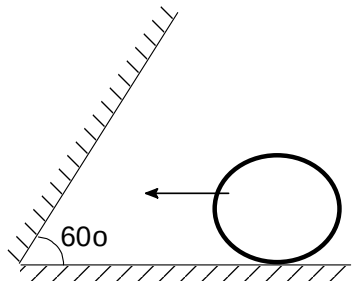


17. The coefficient of restitution between particle and sphere is

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$

(C) $\frac{1}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2}$

18. Above sphere moves towards left and collides elastically with smooth inclined plane having angle of inclination 60°. Then, find Impulse of normal force applied by fixed inclined surface on sphere during collision.



(A) $\frac{80}{\sqrt{3}} N\,S$ (B) $\frac{80}{3} N\,S$

(C) $\frac{20}{3} N\,S$ (D) $80\sqrt{3}\ N\,S$

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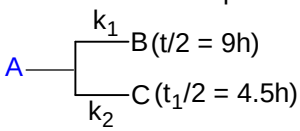
Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. A radioactive isotope A undergoes simultaneous decay to yield B and C nuclei as shown

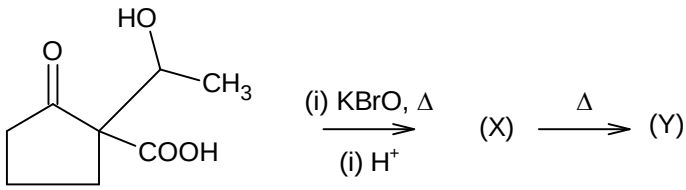


Assume antially no amount of B or C was present, after how many hours will the amount of C be just double to amount of A remaining

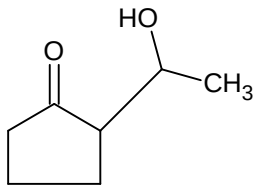
(A) 6 h
(B) 2 hr
(C) 8 hr
(D) 12 hr

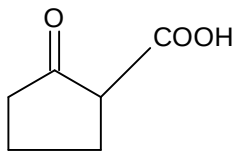
20. The standard reduction potential of metal sparingly soluble salt (AgCl) is 0.209 V and for silver electrode is 0.80 V. If the moles of AgCl that can dissolve in 10 L of a 0.01M NaCl solution is represented as 10^{-z} then z is

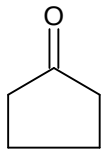
(A) 6
(B) 7
(C) 8
(D) 14

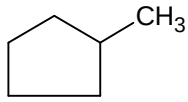
21. 

Product Y is

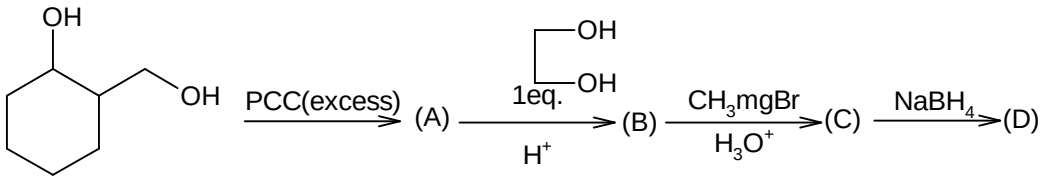
(A) 

(B) 

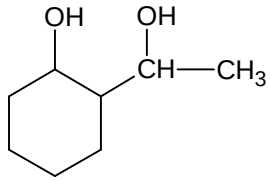
(C) 

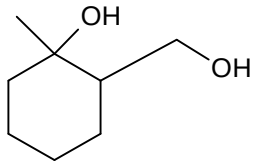
(D) 

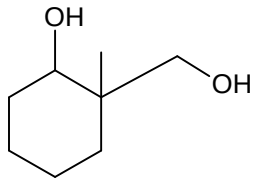
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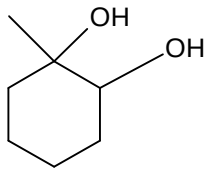
22. 

Product D is

(A) 

(B) 

(C) 

(D) 

23. Two butter (x) and y of ph 4 & 6 are prepared form HA and the salt NaA. Both the buffer are 0.5 M in HA. What would be the pH of the solution obtained by mixing equal volume of the buffer Ka(HA) = 10^{-5}
(A) 4.7 (B) 5.7
(C) 6.7 (D) 8.7

24. The first and second ionization potential of an element M (atomic weight = 25) are 800 and 1500 kj/mol. Calculate the percentage of M^{+2}(g) ions formed if 5g of M absorbs 250 kj of energy.
(A) 70 (B) 40
(C) 60 (D) 30

25. Ballons of 4L capacities are to be filled with hydrogen at a pressure of 1atm and 27^0C from 8L cylinder containing hydrogen at 10 atm at the same temperature. The no of ballons that can the filled is
(A) 20 (B) 18
(C) 40 (D) 38

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. The correct statement regarding adiabatic process is/are
(A) In case of reversible adiabatic expansion of ideal gas $\Delta S_{system} = 0$
(B) In case of reversible ideal gas comparession Δ Ssystem = 0 and Δ Ssorrounding = 0
(C) final temp in irreversible adiabatic process is more as compared to reversible process for both expansion and compression of ideal gas. Initial temperature is kept constant
(D) In adiabatic compression Δu decreases

27. The total pressure is of a binary solution is given by $P = (100 \times A + 260 \times B)\ mm\ Hg$ where X_A and X_B are mole fraction of A and B choose correct statement
(A) vapour pressure of solution is less than the pure B component
(B) vapour pressure of solution is more than that of pure A component
(C) vapour pressure of pure A is 100 mm Hg and that of pure B is 260 mm Hg
(D) the vapour pressure of pure A and B are 260mm Hg and 100mm Hg respectively

28. A mixture contains H atom and Li^{2+} ions. Electron is H-atom is in 3rd excited state and e in Li^{2+} is in 5th excited state. Then which is /are true about the spectral lines observed for the mixture
(A) emitted radiation will contain 21 lines
(B) emitted radiation will contain 19 lines
(C) more than 10 lines are in u.v range
(D) if sample contains only H atom then emission spectrum contains two visible lines

29. Which can form square planar complex
(A) Pt(II), Pd(II) with strong and weak field ligand
(B) Au(III) with strong and weak field ligand
(C) Ni(II) with CN = 4 and strong field ligand
(D) Hg(II) with CN = 4 and strong field ligand

30. For which of the following metal carbon reduction cannot be used
(A) Pb (B) Mn
(C) W (D) Fe

Space for Rough work

31. Select the correct statement
(A) The fraction of edge not occupied by atom in BCC lattice is 0.139
(B) The next nearest neighbour in BCC lattice is 6
(C) Void fraction in ABCABC type structure is 0.32
(D) Distance between two consecutive layer in ccp is $2\sqrt{\frac{2}{3}}R$

32. Identify the correct statement
(A) Hybridisation of carbon in C_3O_2 is sp
(B) $N(Me)_3$ and $N(SiMe_3)_3$ are isostructural
(C) In TBP arrangement ione pair of $\bar{e}$ are usually occupied at equatorial position
(D) Dipole moment of CH_3F is greater than that of CH_3Cl

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 33 and 34

A quantity of 25 mL of solution containing both Fe^{2+} and Fe^{3+} ions is titrated with 25 mL of 0.0200M $KMnO_4$ 2n dilute H_2SO_4. As a result all the Fe^{2+} ions are oxidiced to Fe^{3+}ions. Next 25 mL of the original solution is treated with zn metal. Finally, the solution requires 40 mL of the same $KMnO_4$ solution for oxidation to Fe^{3+}.

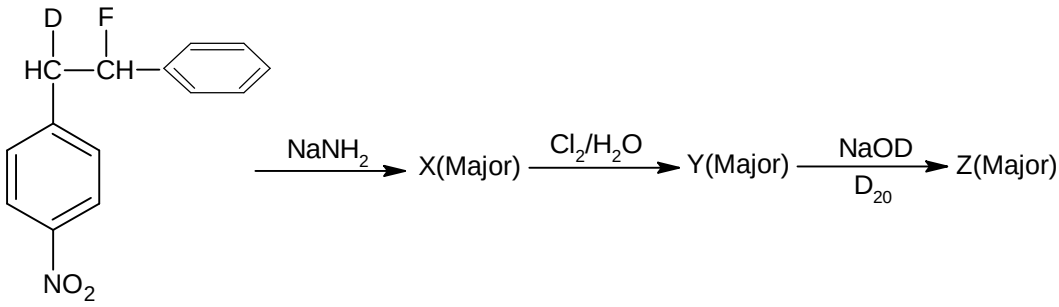
$$MnO4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

33. Molar concentration of Fe^{2+} in the original solution is
(A) 0.01 M (B) 0.02 M
(C) 0.10 M (D) 0.20 M

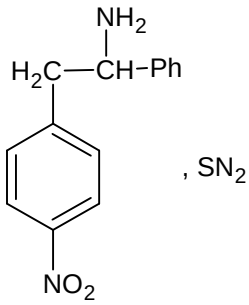
34. Molar concentration of Fe^{3+} in original solution is
(A) 0.06 M (B) 0.16 M
(C) 0.032 M (D) 0.012 M

Space for Rough work

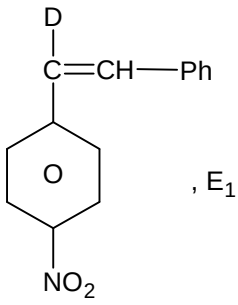
Paragraph for Question Nos. 35 and 36



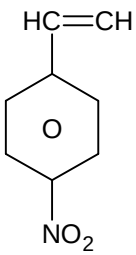
35. The product X and the most possible mechanism of its formation are

(A) 

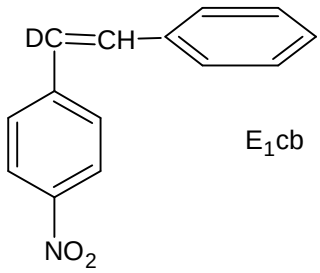
, SN_2

(B) 

, E_1

(C) 

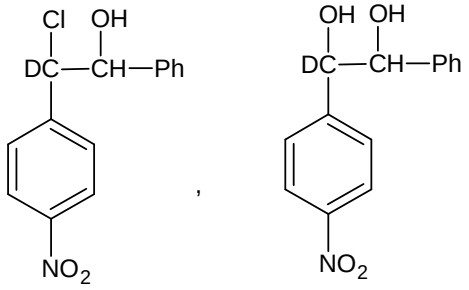
, E_2

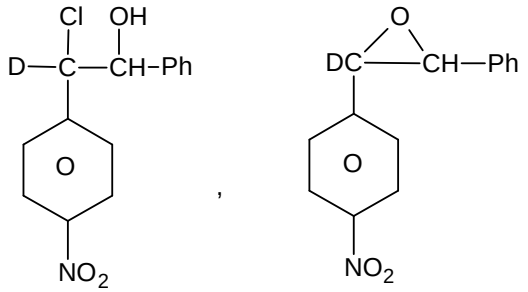
(D) 

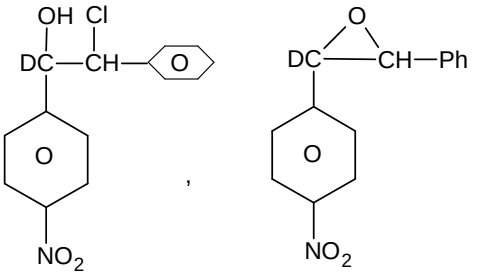
E_1cb

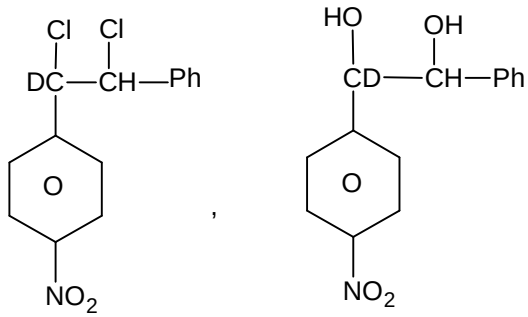
Space for Rough work

36. Product (Y) and (Z) are

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If ω is any complex number such that $z\omega = |z|^2$ and $|z-\bar{z}|+|\omega+\bar{\omega}|=4$, then as ω varies, the area of locus of z is
(A) 4 sq. units (B) 8 sq. units
(C) 16 sq. units (D) 12 sq. units

38. If $\rho \neq 1$ is an nth root of unity, then $1 + 5\rho + 9\rho^2 + \ldots\ldots + (4n-3)\rho^{n-1} = \dfrac{an}{\rho-1}$
and deduce the values of the following.
$$1+5\cos\left(\frac{2\pi}{n}\right)+9\cos\left(\frac{4\pi}{n}\right)+.....+(4n-3)\cos\left(\frac{2(n-1)\pi}{n}\right)=bn$$
$$5\sin\left(\frac{2\pi}{n}\right)+9\sin\left(\frac{4\pi}{n}\right)+.....+(4n-3)\sin\left(\frac{2(n-1)\pi}{n}\right)=c\ n\cot\left(\frac{\pi}{n}\right)$$
for $n \geq 2$, then find the value of a – b – c.
(A) 5 (B) 6
(C) 7 (D) 8

39. Let $I_1=\int_1^{10^4}\frac{\{\sqrt{x}\}}{\sqrt{x}}dx$ and $I_2=\int_1^{10} x\{x^2\}dx$ where {.} denotes fractional part of x. Then
(A) $I_1 = I_2$ (B) $I_1 < I_2$
(C) $I_1 = 4I_2$ (D) $I_1 = 200$

40. Let $f(x)=\int_0^x\left[(a-1)(t^2+t+1)^2-(a+1)(t^4+t^2+1)\right]dt$. Then the value of 'a' for which
$f'(x) = 0$ has two distinct real roots are
(A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-2, \infty)$
(C) $(-2, 2)$ (D) $(-\infty, 2)$

Space for Rough work

41. Let f(x) = min $\left(\ell n\left(\tan x\right), \ell n\left(\cot x\right)\right)$, which of the following statement are incorrect

(A) f(x) is continuous for $x \in \left(0, \frac{\pi}{2}\right)$

(B) Lagrange's mean value theorem is applicable on f(x) for $x \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right]$

(C) Rolle's theorem is not applicable on f(x) for $x \in \left[\frac{\pi}{4}, \frac{3\pi}{8}\right]$

(D) Rolle's theorem is applicable on f(x) for $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$

42. Let X be the set of three digit numbers, which when divided by its sum of its digits give maximum value and Y be the set of all possible real values of a for which the $x^3 - 3ax^2 + 3(298a + 299)x - 2 = 0$ have a positive point of maxima, then the number of elements in $X \cap Y$, is

(A) 0 (B) 6

(C) 7 (D) 9

43. The domain of the function f(x) = $\sqrt{\log_{\sin x + \cos x}\left(|\cos x| + \cos x\right)}$, $0 \le x \le \pi$ is

(A) $(0, \pi)$ (B) $\left(0, \frac{\pi}{2}\right)$

(C) $\left(0, \frac{\pi}{3}\right)$ (D) none of these

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

44. If the global maximum value of $f(x) = -x^2 + ax - a^2 - 2a - 2$ for $x \in [0, 1]$ be '- 5', then 'a' can take the value

(A) $\frac{-4 + 2\sqrt{13}}{3}$ (B) $\frac{-4 - 2\sqrt{13}}{3}$

(C) 1 (D) –3

Space for Rough work

45. A tangent is drawn to the curve y = f(x) at P(x, y), cuts the x-axis at A. Now a perpendicular drawn from P(x, y) to x-axis meets the x-axis at B. If B is mid-point of OA (O being origin) and f(1) = 1, then

(A) general point on the curve is $\left(t, \frac{1}{t}\right)$, where t is a real parameter ($\neq 0$)

(B) y = f(x) is a circle

(C) tangent at P(1, 1) is x + y = 2

(D) f ′(2) = –1/4

46. One vertex of the triangle of maximum area that can be inscribed in the curve $|z-2i|=2$ is $2+2i$, remaining vertices are

(A) $-1+i(2+\sqrt{3})$ (B) $-1-i(2+\sqrt{3})$

(C) $-1+i(2-\sqrt{3})$ (D) $-1-i(2-\sqrt{3})$

47. For $f(x)=\left[\log\left(\sin^{-1}\left(\sqrt{x^2+3x+2}\right)\right)\right]$ ([.] G.I.F.) choose the correct one is

(A) Domain $\left[\frac{-3-\sqrt{5}}{2}, -2\right) \cup \left(-1, \frac{-3+\sqrt{5}}{2}\right]$

(B) Domain $\left[\frac{-3+\sqrt{5}}{2}, -1\right)$

(C) Range is set of all non positive integers

(D) Range is set of all positive integers

48. The co-ordinates of the point of inter section of the curves $y=\cos x, y=\sin 3x, \ -\pi/2 \le x \le \frac{\pi}{2}$.

(A) $\left(\frac{\pi}{8}, \cos\frac{\pi}{8}\right)$ (B) $\left(\frac{\pi}{4}, -\cos\frac{\pi}{4}\right)$

(C) $\left(\frac{\pi}{4}, \cos\frac{\pi}{4}\right)$ (D) $\left(-\frac{3\pi}{8}, \cos\frac{3\pi}{8}\right)$

Space for Rough work

49. Consider the inequality $(x + 3)^n (x – 2) < 0$, $x \in N$. Then the correct statement(s) is/are
(A) the largest integral x satisfying the inequality is 1, if n is even
(B) the least integral x satisfying the inequality is –2, if n is odd
(C) the number of integral x satisfying the inequality is 3, if n is odd
(D) the number of positive integral x satisfying the inequality is 1, if n is even

50. The equation $|x^2 – 1| = a – 1$ can have real solutions for x if a lies in
(A) $[1, \infty)$ (B) $(-\infty, 0)$
(C) $[0, 2]$ (D) none of these

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

Consider a hyperbola $xy = 4$ and a line $y + 2x = 4$. O is centre of hyperbola. Tangent at any point P of hyperbola intersect the co-ordinate axis at A and B.

51. Area of ΔOAB is
(A) Constant and equal to 16 (B) Constant and equal to 32
(C) Constant and equal to 64 (D) Variable

52. Locus of circumcentre of ΔOAB is
(A) An ellipse with eccentricity $\frac{1}{\sqrt{2}}$ (B) an ellipse with eccentricity $\frac{1}{\sqrt{3}}$
(C) a hyperbola with eccentricity $\sqrt{2}$ (D) a circle of radius $\sqrt{2}$

Space for Rough work

Paragraph for Question Nos. 53 and 54

Let t be a real number satisfying
$2t^3 - 9t^2 + 30 - a = 0$ . . . (1)

And $x + \frac{1}{x} = t$ . . . (2)

53. If equation (1) has three real and distinct roots then
(A) $a > 30$ (B) $a < 3$
(C) $3 < a < 30$ (D) $a < 3$ or $a > 30$

54. If $x + \frac{1}{x} = t$ gives six real and distinct values of x, then
(A) $3 < a < 30$ (B) $a \in \phi$
(C) $a \in (2, 5)$ (D) none of these

Space for Rough work